

Riley Brown

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EDUCATION

University of Nebraska-Lincoln

B.S. Mathematics and Data Science, Minor in Computer Science, Lincoln, NE | Aug 2023 - May 2027 (expected)

- GPA 3.92/4.00, University Honors Program

PREPRINTS

Fixed-Parameter Tractability of t -Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv preprint, 2026

arXiv

RESEARCH EXPERIENCE

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln, Lincoln, NE | Fall 2026

- Testing whether collision-free geodesic-flow operators on Ramanujan complexes can sharpen iterative-decoder thresholds for quantum LDPC codes

Research Assistant, pseudorandomness

North Carolina State University, Raleigh, NC | Summer 2026

- Studied Ta-Shma's epsilon-balanced code construction with Prof. Chin Ho Lee as a research partner
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zheng construction

Research Assistant, t -uniform hypergraph degree sequences and FPT algorithms

Budapest Semesters in Mathematics, Budapest, Hungary | Spring 2026

- Proved t -uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher, algorithm design

Harvey Mudd College, Claremont, CA | Summer 2025

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics (team of four)
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

PROJECTS

leanverify

Lean 4 proof checker against Mathlib: rejects sorries and smuggled axioms | python, lean 4 | 2026

SKILLS

AI for mathematics: Lean 4 and Mathlib verification tooling, LLM-assisted proof search, agentic research harnesses

GRADUATE COURSEWORK

Bijjective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Algebra I, Coding Theory, Development and Analysis of Algorithms

ACTIVITIES

President, Pi Mu Epsilon; President, Math Club; Directed Reading Program (combinatorial game theory); Putnam Exam Practice; Quiz Bowl Club; Poker Club